

PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

1. RESOURCE MANAGEMENT AND ARCHITECTURAL ROLES DISTRIBUTION

The architecture, integration, and delivery validation lifecycles were successfully executed according to engineering roles:

- * Core Machine Learning & Data Pipeline Logic: Managed data cleaning routines, train-test splitting metrics, Random Forest ensemble hyperparameter tuning, and model pickle exports.
- * Web Integration Backend Framework: Designed Flask controller loops, set up POST route handlers, and parsed client string payloads into numerical arrays.
- * Frontend Presentation Layouts: Built responsive HTML interfaces using Bootstrap v5 responsive column utilities and stylized alert tags.
- * Quality Validation & Compilation: Supervised technical functional analysis testing loops and compiled documentation specifications.